

PETER A. SAYEGH

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EDUCATION

Columbia University | Columbia Engineering

New York, NY

MS in Electrical Engineering – Nikola Tesla Scholarship, GPA: 3.79/4.0

Expected Dec 2026

Coursework: Digital Signal Processing, Adv. Analog Integrated Circuits, Power Electronics, MOS Transistors, Analog-Digital Interfaces in VLSI, Embedded AI, Microwave Circuit Design, Motor Drive Systems, High-Frequency Magnetics Design

Activities: Lead Department Representative for Electrical Engineering within the Engineering Student Graduate Council (EGSC)

École Polytechnique

Palaiseau, FR

BS in Mathematics and Physics, GPA: 3.75/4.0

Jun 2025

Coursework: Convex Optimization, Stochastic Processes, Electrodynamics, Adv. Quantum Physics, Thermodynamics, Statistical Physics, Condensed Matter, Algorithms, Numerical Analysis, Web Programming, Object-Oriented Programming, Machine Learning

EXPERIENCE

Columbia Engineering

New York, NY

Course Assistant

Sep 2026 – Present

- Supported Prof. Guido Jajamovich and his students in Digital Signal Processing by holding weekly office hours and review sessions and grading homework, assignments, and projects

Illumination Diagnostics

New York, NY

Electrical Engineering Intern

Jun - Aug 2026

- Developed a heart-rate extraction pipeline sampling 12 optical channels at 31 Hz, achieving 1–3 bpm agreement against reference
- Prototyped PPG and temperature acquisition systems using ESP32 microcontrollers with synchronized respiration belt recording
- Assembled a 940 nm IR LED illumination array on perfboard (~14 mA/LED) to provide stable background illumination
- Designed a 2-axis pan/tilt laser targeting system (dual servos, Arduino + Waveshare driver), achieving $<0.1^\circ$ angular resolution for programmable wall-target positioning

ELECTRICAL ENGINEERING PROJECTS

Columbia Engineering

New York, NY

Fully Differential Switched-Capacitor Amplifier

Feb - May 2026

- Designed folded-cascode OTA with switched-capacitor CMFB in $0.18\ \mu\text{m}$ CMOS: 56.6 dB gain, 69° phase margin, 487 μW
- Met 0.09% settling error in a 28 ns window for a 7-bit 16 MHz pipelined ADC, at 0.24 pJ/step

Columbia Engineering

New York, NY

250 MS/s 8-bit Current-Steering DAC

Feb - May 2026

- Designed a 3+5 segmented current-steering DAC in $0.18\ \mu\text{m}$ CMOS: $1.53\ V_{\text{pp,diff}}$ into $100\ \Omega$ at 250 MS/s, 27.6 mW
- Sized 255 unit cells to 0.67% $\sigma(I)/I$ mismatch, bounding 3σ DNL under 0.65 LSB; SFDR ≥ 60 dB across corners

Columbia Engineering

New York, NY

SignSpeak: ASL Recognition on Microcontroller

Feb - May 2026

- Deployed 16K-parameter depthwise-separable CNN on ESP32S3 via TFLite Micro INT8 quantization, achieving 97.4% accuracy in 67.4 KB flash
- Built a full on-device inference pipeline from camera capture to OLED display, <110 ms latency with no cloud dependency

RESEARCH

École Polytechnique - Center for Theoretical Physics

Palaiseau, FR

Levitation and Stability in Maglev Systems

Feb 2024 - Jul 2026

- Derived a closed-form dipole-layer force model for a scaled maglev prototype, fitting 24 gap-versus-load measurements with $R^2 = 0.96$ and predicting the unloaded levitation gap within 0.4 mm
- Extrapolated to the Transrapid TR08 using a single fitted parameter, reproducing its 574-passenger nominal capacity
- First-author manuscript provisionally accepted, European Journal of Physics

TECHNICAL SKILLS

CAD Tools: Cadence Spectre, Virtuoso, ADE, PLECS, LTSpice, Keysight ADS, Xschem, ngspice, KLayout, Ansys Maxwell

Lab and Test: Oscilloscope, function generator, DMM, soldering and rework, perfboard prototyping, benchtop characterization

Programming Languages: Python, C++, Arduino, R, MATLAB, HTML, CSS, JavaScript, QML

Data Science and Machine Learning: TensorFlow, Keras, JAX/Optax, scikit-learn, NumPy, SciPy, pandas, statsmodels, DSPy

Scientific Computing: QuTiP, Scikit-HEP, PySpice

Languages: French (Native), Arabic (Native), German (Advanced), Spanish (Intermediate)